

# K-means and DBSCAN: Performances linked to Data Topology Dependence

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*Clustering effectiveness strongly depends on the intrinsic geometric and density properties of the dataset. In this work, we evaluate K-means and DBSCAN across four synthetic scenarios: high-dimensional non-convex structures, anomaly detection in autoregressive time series, shape-based pattern extraction, and a non-linearly separable checkerboard topology. Dimensionality reduction techniques and quantitative metrics are employed to visualize and validate performances. Our results demonstrate that standard algorithms succeed only when their underlying structural assumptions are met, but fail significantly on non-linearly separable or uniformly dense topologies. This underscores the critical necessity of tailoring clustering techniques to the intrinsic properties of the data.*

## INTRODUCTION

The aim of this assignment is to investigate the performance of different clustering techniques in a variety of unsupervised learning problems. In particular, we analyze how the characteristics of a dataset influence the effectiveness of K-means and DBSCAN. To this end, we consider four distinct problems that highlight the strengths and limitations of these clustering methods.

In the **first problem**, the objective is to identify clusters in a 12-dimensional dataset. Since direct visualization of such space is impossible, we first compare different dimensionality reduction techniques that allow for a 2-dimensional representation of the data. We then evaluate the performance of several clustering algorithms.

The **second problem** focuses on anomaly detection in a collection of time series of equal length. The goal is to isolate the anomalous series by identifying it as an outlier with respect to the remaining data, which are expected to form a single cluster.

In the **third problem**, we analyze another time series collection containing different hidden patterns embedded in noise. The objective is to determine whether clustering techniques can distinguish time series containing sinusoidal or square-wave signals from those containing noise only.

Finally, in the **fourth problem**, we analyze a dataset that highlights the limits of the clustering techniques used.

## DATA GENERATION

This section outlines the data generation process for the four problems. Generated labels are retained solely for evaluating clustering performance.

### Problem 1

For this experiment, we generate 600 3-dimensional data points, laying on different parametric curves: class 0 lies on a ring-like curve and is composed of 35% of the data, class 1 on three translated ellipses and is composed of 45% of the data, and class 2 on two symmetric oblique

lines, and is composed of 20% of the data. Gaussian noise is added to each point. The data is then embedded in a 12-dimensional space by appending nine additional noise dimensions. Finally, the data is rotated with a random orthogonal matrix in order to hide the underlying structure. The original 3-dimensional data topology is shown in Figure 1.

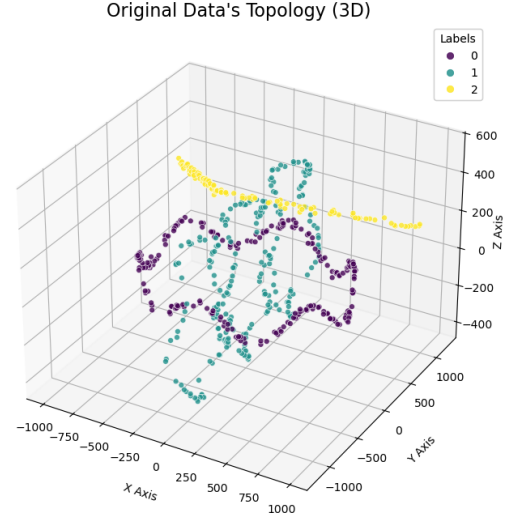


FIG. 1: 3D data for Problem 1, before the addition of noise components to embed them in a 12D space.

### Problem 2

The data for this experiment are 101 time series of length 100. They are generated as autoregressive processes of first order AR(1):

$$x_t = \phi \cdot x_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

with a recall force parameter  $\phi = 0.8$  and white noise standard deviation  $\sigma = 5$  [1]. Every time series can be interpreted as a point in  $\mathbb{R}^{100}$ . An anomaly is then introduced in one of the time series by adding a constant offset whose amplitude varies within  $\{5, \dots, 25\}$  with unitary increments.

### Problem 3

The data for this problem consist of 120 time series generated as AR(1). A third of the series are augmented with a sinusoidal pattern, another third with a square-wave pattern, while the remaining contain only the AR(1) generated noise. The obtained set of time series is shown in Figure 2.

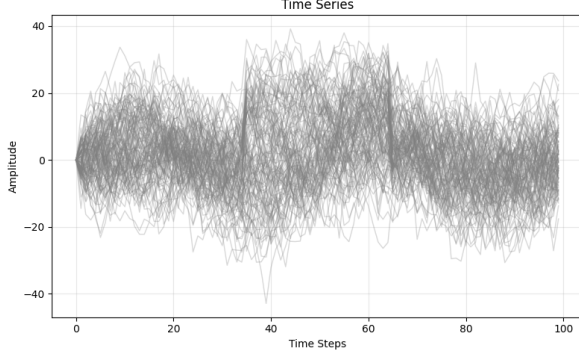


FIG. 2: Dataset for problem 3: set of 120 time series with different anomalies.

### Problem 4

In this problem the data are 2-dimensional with binary labels, disposed in a checkerboard structure over a section of the plane as depicted in Figure 3.

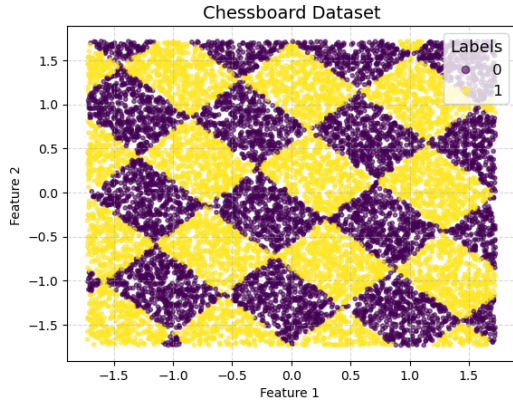


FIG. 3: Labeled dataset for Problem 4.

## METHODS

In this section we illustrate the methods used for data visualization, clustering, and performance evaluation.

### Visualization Methods

For data visualization, three dimensionality reduction methods are considered:

**Principal Component Analysis (PCA):** a linear dimensionality reduction technique that projects data onto a new orthogonal basis formed by a subset of  $k$  eigenvectors of the sample covariance matrix. The selected eigenvectors maximize the variance retained in the lower-dimensional representation [2, 3].

**t-Distributed Stochastic Neighbor Embedding (t-SNE):** a non-linear dimensionality reduction technique that preserves the local structure of data by modeling pairwise distances as similarity probabilities. The low-dimensional embedding is optimized by minimizing the Kullback-Leibler divergence between these high-dimensional similarities and a heavy-tailed Student's t-distribution, which allows well-separated clusters to emerge without crowding distant points in the center [3].

**Metric Multidimensional Scaling (Metric MDS):** a geometry-based dimensionality reduction technique that seeks a low-dimensional embedding by preserving global pairwise distances. It learns the new coordinates by minimizing a stress function, which penalizes the squared differences between the spatial distances in the original high-dimensional space and those in the lower-dimensional representation. The optimization process is governed by the target dimensionality, the number of random initializations  $n_{init}$ , and the maximum number of iterations  $max_{iter}$  [3].

### Clustering Algorithms

Clustering aims to partition a dataset into groups of similar points based on a defined notion of similarity. In this work, three widely used clustering algorithms are considered:

**K-means:** a centroid-based algorithm that partitions the dataset into  $K$  clusters by minimizing a cost function that measures the total sum of squared distances between each data point and the centroid of its assigned cluster. The algorithm iteratively updates both the clusters  $C_j$  and their centroids  $\mu_j$  for all  $j \in \{1, \dots, K\}$  until convergence [2, 3].

**Density-Based Spatial Clustering of Applications with Noise (DBSCAN):** a density-based algorithm that groups closely packed points while identifying those in sparse regions as outliers. This approach allows it to discover clusters of arbitrary shape without predefining their number. The algorithm relies on two key parameters:  $\varepsilon$ , the maximum distance for two points to be considered neighbors, and  $m_s$ , the minimum number of points required within this  $\varepsilon$ -radius to form a cluster [3].

**Hierarchical Clustering:** builds a nested sequence of clusters by iteratively merging smaller groups based on a defined linkage criterion. The resulting hierarchical structure is visualized using a *dendrogram*, a tree-like diagram that records the distance or dissimilarity at which

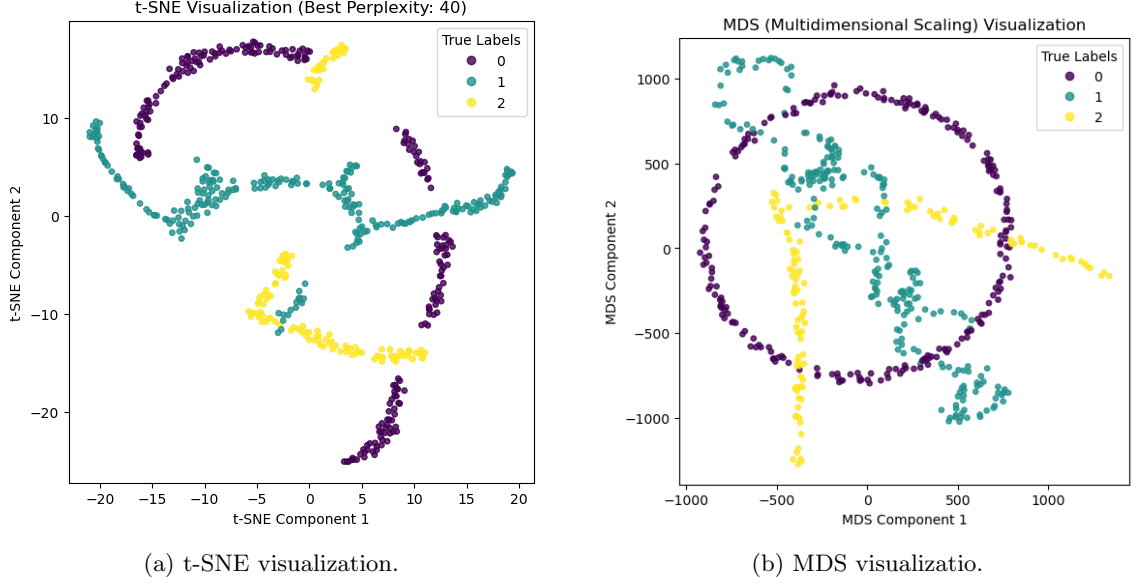


FIG. 4: Comparison of t-SNE and MDS reduction methods. The plots are colored according to the true labels.

each merge occurs. This approach allows for the determination of  $K$  by drawing a horizontal cut across the dendrogram at a specific height, defining the clusters based on a desired distance threshold [3].

### Performance Metrics

Finally, we implement several metrics to assess the quality of the clustering:

**Normalized Mutual Information (NMI):** used to quantify the information shared between the clusters identified by the algorithm and the ground truth labels provided in the dataset. It can be evaluated as

$$\text{NMI}(Y, C) = \frac{2I(Y; C)}{H(Y) + H(C)}, \quad (2)$$

where  $I$  represents the mutual information between real labels  $Y$  and predicted ones  $C$ . The normalization factor is given by the sum of the entropies  $H$  associated to both  $Y$  and  $C$  [4]. NMI values range from 0 to 1, with one indicating that the clusters perfectly correspond to the known classes and zero indicating no correlation.

**Silhouette Index:** an evaluation metric that measures how well each point is assigned to its cluster by comparing intra-cluster cohesion with inter-cluster separation. For each data point, the silhouette value is defined as:

$$s = \frac{b - a}{\max(a, b)}, \quad (3)$$

where  $a$  is the average intra-cluster distance and  $b$  is the mean distance to the nearest neighboring cluster. The overall index is the average  $s$  across all points. Scores

approach 1 for well-separated and cohesive clusters, fall near 0 for overlapping clusters, and become negative in cases of likely misclassification [3].

**F1 Score:** a metric that evaluates clustering performance with respect to ground truth classes by combining precision and recall. Precision measures the fraction of correctly assigned points within a cluster, while recall measures the fraction of a true class successfully captured. They are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (4)$$

where  $TP$ ,  $FP$ , and  $FN$  represent True Positives, False Positives, and False Negatives, respectively [5]. The F1 score is their harmonic mean; high values indicate an accurate balance between capturing all relevant class members and excluding irrelevant points [2, 3].

Algorithms and metrics are implemented using the Python package **Scikit-Learn** [6].

## RESULTS

### Problem 1

To visualize the original 3D data structure, PCA, t-SNE, and MDS were evaluated. Since PCA requires multiple projection planes to fully capture the topology, compact 2D representations were favored. For t-SNE, a grid search maximizing the silhouette score yielded an optimal perplexity of  $p_{best} = 40$ . For MDS, the parameters were set to  $n_d = 2$  dimensions,  $n_{init} = 2$  initializations to optimize runtime without visual quality loss, and  $max_{iter} = 200$ . Ultimately, MDS was selected to present

the results because, similarly to PCA, it preserves global spatial patterns without introducing the structural distortions typical of t-SNE, as shown in Figure 4.

With the spatial distribution and global structure of the dataset clarified through this preliminary analysis, the data was subsequently partitioned. The first clustering algorithm considered was K-means.

To determine the optimal number of clusters  $K$ , hierarchical clustering was performed, and the optimal value was selected based on the largest jump in the dendrogram, as shown in Figure 5.

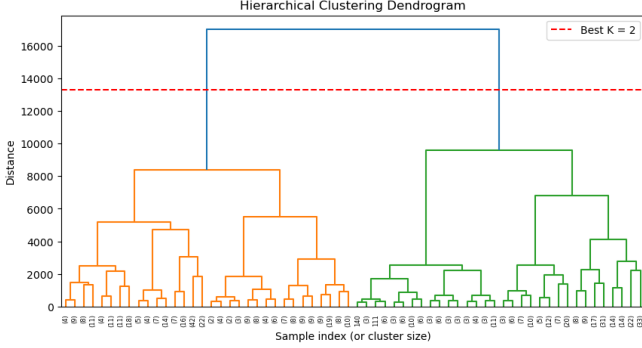


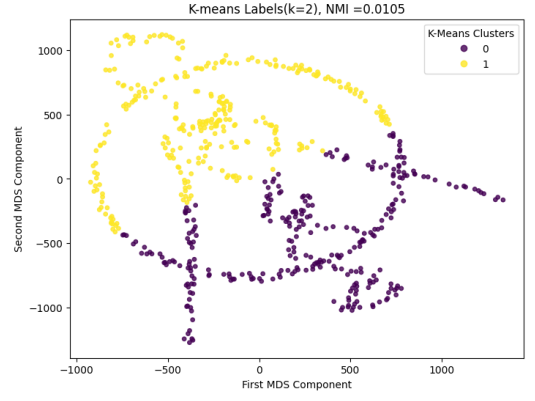
FIG. 5: The dendrogram resulting from hierarchical clustering in Problem 1.

The best value for the  $K$  parameter was found to be  $K = 2$ . With an NMI score of 0.0105, the failure of this algorithm is clear. This outcome was expected, as K-means assumes convex, linearly separable clusters, an assumption violated by the present dataset. Next, the DBSCAN algorithm was applied. The minimum sample parameter  $m_s$  was set to 3, as  $m_s = 1$  would reduce each data point to a core point, eliminating the density requirement, and  $m_s = 2$  effectively reduces DBSCAN to hierarchical clustering with single linkage. The  $\varepsilon$  parameter was estimated by identifying the point of maximum second derivative of the  $k$ -NN distance curve, after applying Savitzky–Golay smoothing [7].

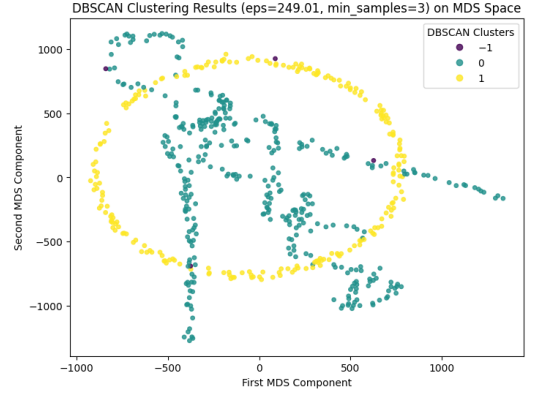
Resulting parameters were

$$(\varepsilon, m_s) = (249.01, 3). \quad (5)$$

This value of  $\varepsilon$  is consistent with the result of a grid search over the NMI, which yielded  $\varepsilon_{g-s} = 255.0$ . While the algorithm successfully identifies the cluster corresponding to the ring-like curve, it merges the remaining two clusters, likely prioritizing density over geometric shape. Another four points are identified as outliers. The resulting NMI score of 0.7527 represents a substantial improvement over K-means, though this is still not a satisfying result, since the algorithm failed to correctly recognise all three groups lying in the data. The graphical results of both algorithms are shown in Figure 6.



(a) K-means results.



(b) DBSCAN results.

FIG. 6: Results of algorithms used in Problem 1.

## Problem 2

To address Problem 2, DBSCAN was preferred over K-means. Since K-means minimizes within-cluster variance, initializing it with  $K = 2$  forces it to arbitrarily bisect the dense cloud of normal time series to balance the variance, rather than isolating the single anomaly, which would result in a disproportionately high cost function. DBSCAN avoids this by evaluating Euclidean distances in the 100-dimensional space. The parameter  $m_s$  was set strictly greater than 1, as  $m_s = 1$  would cause the anomaly to merge with the main cluster.

To estimate the optimal  $\varepsilon$ , the pairwise squared Euclidean distances  $D^2 = \sum_{t=1}^L (X_t - Y_t)^2$  were analyzed, yielding an empirical mean distance of  $d_{mean} \approx 113$ . This aligns with the theoretical expectation for independent AR(1) processes [1].

Given the stationary variance  $\sigma_t^2$ , the expected distance between two regular trajectories is:

$$d_{mean} = \sigma \sqrt{\frac{2L}{1 - \phi^2}} \quad \text{with} \quad \sigma_t^2 = \frac{\sigma^2}{1 - \phi^2}. \quad (6)$$

The radius was set to  $\varepsilon = 130$ , adding a margin to

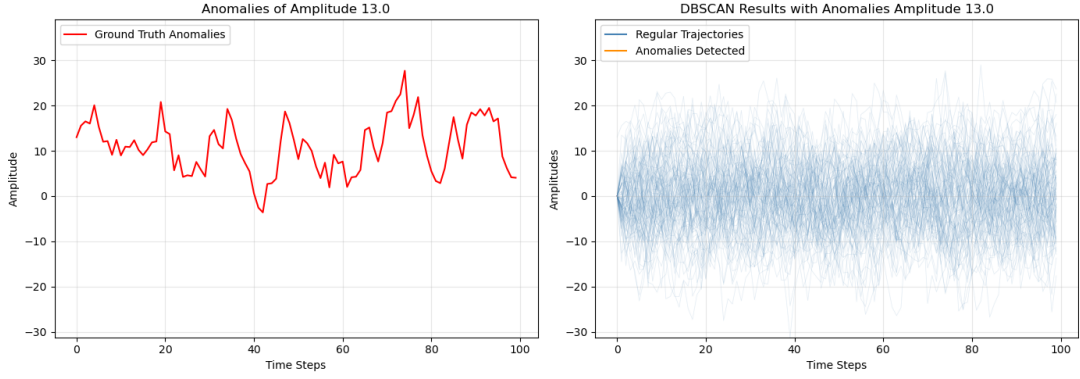
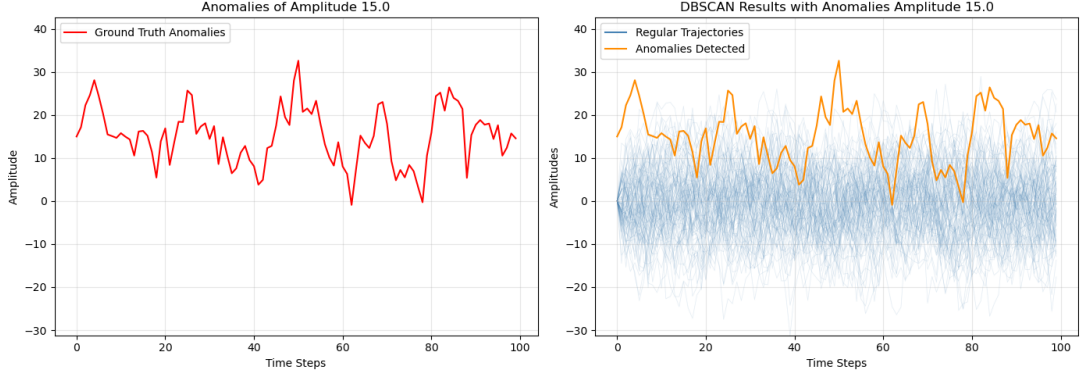
(a) Analysis with injected anomaly of amplitude  $A = 13$ .(b) Analysis with injected anomaly of amplitude  $A = 15$ .

FIG. 7: DBSCAN evaluation with respect to two different injected anomalies.

strictly isolate the anomaly while minimizing both false positives and false negatives.

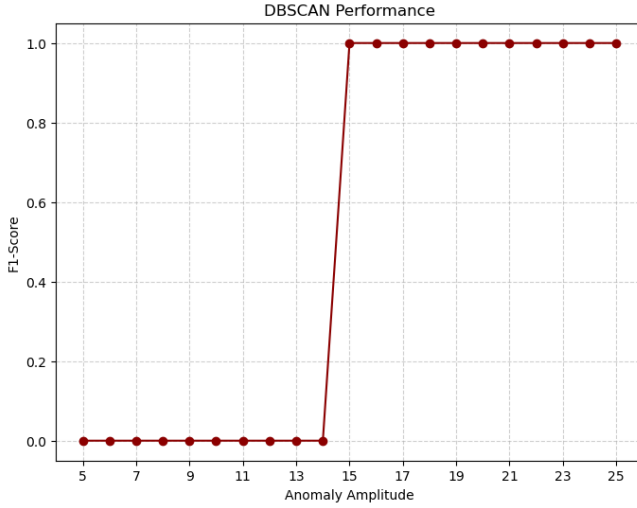


FIG. 8: F1 score of DBSCAN for different offset amplitudes of the injected anomaly.

The algorithm was then evaluated across varying offset amplitudes  $A$ . As shown by the F1 scores in Figure 8,

two operational regimes emerge: DBSCAN fails to isolate the anomaly for  $A < 15$ , but successfully detects it for  $A \geq 15$ . Figure 7 illustrates the ground truth anomalies and the corresponding DBSCAN results for two representative amplitudes.

### Problem 3

In order to correctly use K-means algorithm to solve this problem, the best  $K$  parameter is determined.

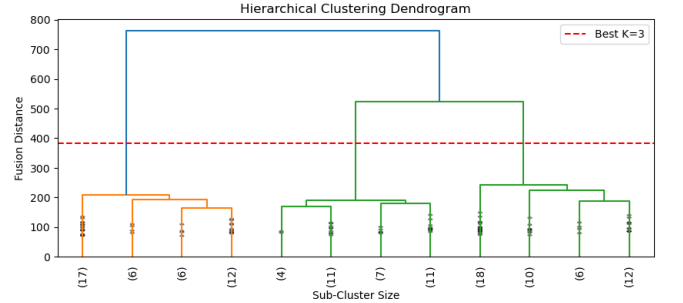


FIG. 9: The dendrogram resulting from hierarchical clustering in Problem 3.



To do so, hierarchical clustering is performed, selecting the optimal value for  $K$  from the maximum jump that occurs in the dendrogram, reported in Figure 9.

This analysis yielded a value of  $K = 3$ , that was then confirmed by a parameter search using the NMI as an evaluation metric. The obtained maximum value is

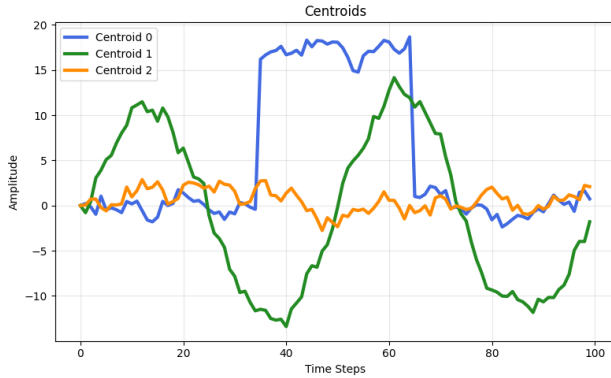
$$\max_K \text{NMI} = \text{NMI}(K = 3) \approx 0.90. \quad (7)$$

It is important to note that this second method makes use of the ground truth labels. Therefore it does not constitute an estimation of  $K$ , but rather a validation of the result obtained from the dendrogram. K-means algorithm was then executed setting  $K = 3$ . The results are reported in Figure 10.

The resulting clusters are depicted in Figure 10a. The centroids reported in Figure 10b confirm that K-means has successfully recovered the underlying patterns: centroid 0 resembles a square wave, centroid 1 a sinusoidal wave, centroid 2 a flat, noisy signal. Thus the three groups defined in the data generation process have been recovered.



(a) Clusters recovered with K-means.

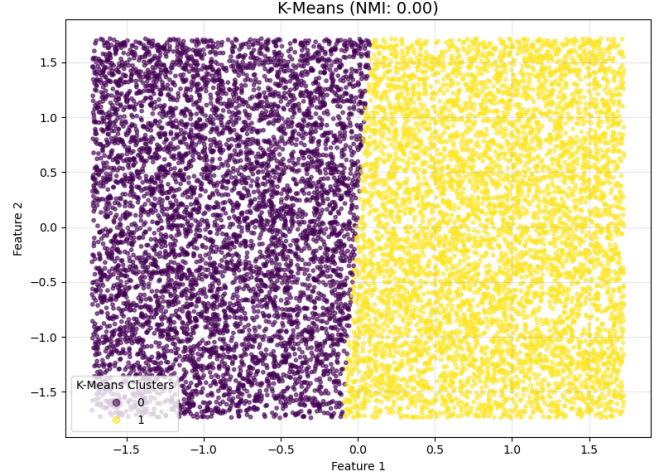


(b) Centroids computed with K-means.

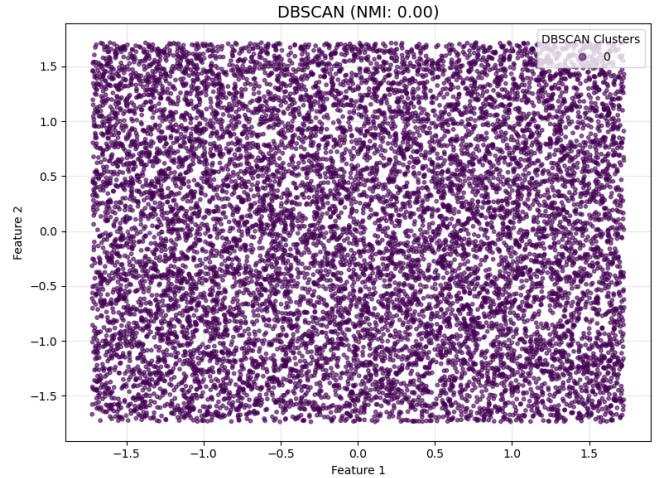
FIG. 10: Results of K-means clustering in Problem 3.

#### Problem 4

To address Problem 4, K-means and DBSCAN were applied to recover the original labels. However, both algorithms proved unsuccessful. K-means failed due to the non-linear separability of the data, since it assumes convex, linearly separable clusters, which does not hold true for this dataset. Conversely, DBSCAN struggled with the dataset's uniform density. As a density-based approach, it merged all data points into a single cluster, without detecting any internal density variations. The results are reported in Figure 11.



(a) K-means results.



(b) DBSCAN results.

FIG. 11: Results of algorithms used in Problem 4.

## CONCLUSIONS

In this assignment, the performance and limitations of DBSCAN and K-means clustering algorithms were compared across different unsupervised learning problems. The results show that no approach is universally optimal, and different problems may require different clustering strategies.

The first problem showed how centroid-based approaches like K-means fail to identify non-convex clusters, while DBSCAN proved far more effective at recovering the underlying topology of the data.

The second problem demonstrated DBSCAN's effectiveness in anomaly detection within time series. By defining an appropriate distance threshold, the algorithm correctly detected the anomalous time series.

Conversely, the third problem highlighted how K-means is more suitable for shape-based clustering, since the algorithm managed to correctly separate the time series based on the hidden patterns.

Finally, the fourth problem exposed the fundamental limitations of both algorithms, highlighting how they struggle with issues like non-linear separability (K-means) and uniformity in spatial density (DBSCAN).

In conclusion, this analysis showed that, in order to correctly address clustering problems, a careful preliminary evaluation of the dataset's geometry, density distribution, and dimensionality is necessary.

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